

Qualitative & quantitative data

Before you can analyse anything, you have to know what kind of data you're holding — and why you bothered to collect it. This is the first decision in every data project, and examiners love it because it's the one students rush.

What this key knowledge covers

This summary distils everything you need for **KK01 — Qualitative & quantitative data** in Data Analysis. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

Words or numbers?

All data splits into two families. The fastest way to tell them apart is to ask one question: **can I count or measure it, or am I describing it?**

- Answers **how many, how much, how often**
- Can be added, averaged, ranked and charted
- Objective and easy to compare at scale
- Weak at explaining the *why* behind a number
- Answers **what kind, why, how does it feel**
- Rich, detailed, captures context and emotion
- Must be read, grouped and interpreted (coded)
- Harder to summarise or compare across many people

A memory hook that actually sticks

The **N** stands for **Numbers**. If you can do maths with it — sum, average, plot it — it's quantitative.

The thermometer & the diary

Imagine a friend has a fever. Two tools tell you different things:

“38.9 °C.” A precise number you can track over hours, compare to normal, and chart. But it can't tell you they feel miserable.

“Aching all over, couldn't focus, just wanted to sleep.” It captures the human experience the number leaves out — but you can't average it.



Neither is “better.” Good data analysis usually **combines both**: the numbers tell you *how much*, the words tell you *what it means and why*. Remember that line — it answers half the exam questions on this dot point.

Become the data source

Theory sticks faster when you generate the data yourself. Two quick tasks: **count** something — you'll produce a number — then **describe** something — you'll produce words. That's the whole distinction, made by your own hands.

Your answer is a **number**. Numbers can be totalled, averaged, ranked and charted — so counting always gives you quantitative data. You just acted like a sensor: measuring the world and outputting a value.

Those are **words, qualities and opinions** — that's qualitative data. Here's the key idea: it's *subjective*. A classmate looking at the very same object might call it *boring* where you said *pretty* — and both answers are valid data. You can't average “pretty”, but it captures meaning and feeling that a number never could.

Did you feel the difference? Counting forced one exact number. Describing opened up many possible words. That's why the two types answer different questions — and why good analysis usually needs both.

Why collect data at all?

The dot point says “types **and purposes**.” Students remember the types and forget the purposes — then lose marks when a question asks them to *justify a choice*. Data is collected so an organisation can do one of these jobs:

Capture what is happening right now — a snapshot of users, behaviour or conditions.

Set groups, times or options side by side to see differences.

Spot trends, clusters or relationships hiding in the raw data.

Understand the *reasons* behind a behaviour or result.

Use what's happened to estimate what's likely next.

Back a recommendation with evidence others will trust.

Type follows purpose

Notice the pattern: if the purpose is to **compare or predict**, you usually want quantitative data you can chart. If the purpose is to **explain or understand why**, you usually need qualitative data. *The purpose chooses the type* — not the other way around.

Six places data comes from

These are the six sources named in the study design. Tap each to see what it produces inside StepUp, whether it's qualitative or quantitative, and the purpose it serves.

A researcher asks open questions and follows up on the answers. The result is rich, detailed and personal — and impossible to reduce to a single number.

The classic trap. A survey can collect *both* types at once. **Closed** questions (multiple choice, Likert scales, yes/no) give quantitative data; **open** questions (“tell us why...”) give qualitative



data.

Devices measure the physical world and output numbers, continuously and without a human typing anything. Step counters, heart-rate monitors, GPS, temperature probes.

Structured facts describing who people are — age, year level, location, household size. The Australian Bureau of Statistics (ABS) Census is the national example. Mostly counted and grouped.

Not a different *kind* of number — a different *shape*. The same value captured at regular intervals (daily, hourly) so you can see how it changes. Time is always one axis.

An AI model produces new data — a written summary (qualitative), a predicted number (quantitative), or a synthetic dataset. Useful, but it's *generated*, so it can be biased, wrong or invented. Always treat it with caution.

→ The **quality** of all this data — accuracy, bias, reliability — is the next dot point (KK03). Here we only classify it and name its purpose.

The StepUp data stream

Every source feeds the same app. Tap a source — on the buttons or the orbiting node itself — to send its data flowing into StepUp's core. Cool / cube tokens are quantitative ; warm / round tokens are qualitative. Sources that produce both emit a mix.

Sort it: words or numbers?

Drag each piece of StepUp data into the right bin — or tap it, then tap a bin on touch screens. Watch the trap cards: a number isn't always quantitative.

Key terms

Data expressed as numbers — countable or measurable. Answers how many / how much.

Data expressed as words, images or descriptions. Answers what kind / why / how it feels.

A question with set responses (yes/no, scale, multiple choice). Produces quantitative data.

A question inviting a free response in the person's own words. Produces qualitative data.

Numeric data generated automatically by hardware measuring the physical world.

The same measure captured at regular intervals so change over time can be seen.

Structured facts describing a population — age, location, year level, etc.

Grouping and labelling free-text responses so qualitative data can be summarised.

Quick quiz

Instant feedback. No marks counted here — it's a warm-up for the real practice below.

P·E·L writing trainer

Short-answer marks come from *structure*, not length. Use **Point · Evidence · Link**: state the type, give a concrete example, link it to the purpose.

Name the data type clearly.

Give a specific example from the scenario.

Tie it to the purpose / why it suits the goal.

[Point] The team should collect qualitative data. [Evidence] For example, they could run short interviews or use an open survey question asking students to describe what made them stop walking. [Link] Qualitative data is suited to this purpose because it captures the reasons and feelings behind behaviour — the “why” — which a step count or rating alone could never explain.

VCE-style questions

Four questions, 15 marks. Try each one on paper first. Then reveal a weak answer, a strong answer and the marking guide — and mark your own work using the dock in the corner.

Qualitative is words and quantitative is numbers. Like steps and feelings.

Quantitative data is data expressed as numbers that can be counted or measured — for example, StepUp recording 8,412 steps in a day. Qualitative data is data expressed as words or descriptions — for example, a student's interview comment that the walking route feels unsafe.

- 1 mark: correct definition + matched example for quantitative (numbers / countable; e.g. step count).
- 1 mark: correct definition + matched example for qualitative (words/description; e.g. interview comment).

They should use a survey to get quantitative data so they can make a graph of how many people stopped.

They should collect qualitative data, using interviews (or open-ended survey questions) with students who stopped. This is appropriate because the purpose is to understand reasons and motivations — the “why” behind the behaviour — and qualitative data captures rich, detailed explanations that a number could not provide.

- 1 mark: identifies qualitative data.
- 1 mark: names a suitable method (interview / open-ended question / focus group).
- 1 mark: justification links the type to the purpose of understanding reasons/the “why”.

Both are qualitative because they are about exercise and feelings. The purpose is to help students.

The whole dot point on one screen

Quantitative = numbers you can count/measure (how many, how much). **Qualitative** = words/descriptions (what kind, why). **Type follows purpose.**

Watch for: numbers used as labels aren't truly quantitative · a survey collects both · always pair type with purpose · All data needs checking.

Common traps & misconceptions

Exam trap

T1 “It's a number, so it must be quantitative.” Not always. A student ID like 047, a postcode, or a jersey number is a **label**, not a measurement. If averaging it is meaningless, it's acting as categorical (qualitative-style) data. **Fix:** ask “can I do meaningful maths on this?” If no, it's not really quantitative.

Exam trap

T2 Treating a survey as one type only. A survey is a **method**, not a data type. The same form can collect both — closed questions give numbers, open questions give words. **Fix:** classify the *question*, not the survey.

Exam trap

T3 Forgetting the “purposes” half. The dot point is types **and purposes**. If a question says “justify” or “explain why,” naming the type alone won't score full marks — you must link it to the job it does. **Fix:** always pair a type with a purpose: “qualitative, *because they need to understand why...*”.

Exam trap

T4 Calling qualitative data “just opinions” / less valid. Both types are legitimate evidence. Qualitative data is the only way to capture the *why*. Dismissing it shows a shallow understanding. **Fix:** frame them as complementary, each suited to different purposes.

Exam trap

T5 Trusting AI-generated data automatically. AI output is generated, not observed. It can be biased, outdated or fabricated. It still counts as data, but it needs verifying. **Fix:** note that AI-generated data must be checked for accuracy and bias before use.

Exam trap

T6 Confusing a Likert scale's feeling with its data. “Rate your motivation 1–5” is *about* a feeling, but the data captured is a **number** — so it's quantitative. **Fix:** classify the recorded value, not the topic it asks about.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Define · 2 marks

Qualitative data and quantitative data. Give one example of each from the StepUp app.

✗ A weak answer

Qualitative is words and quantitative is numbers. Like steps and feelings. True but too thin. No clear definitions, examples aren't tied to a type, and "feelings" isn't a concrete data item. Likely 0–1/2.

✓ A strong answer

Quantitative data is data expressed as numbers that can be counted or measured — for example, StepUp recording 8,412 steps in a day. Qualitative data is data expressed as words or descriptions — for example, a student's interview comment that the walking route feels unsafe. Clear definition of each + a specific StepUp example correctly matched to its type. Full marks.

How the marks are awarded

- 1 mark: correct definition + matched example for quantitative (numbers / countable; e.g. step count).
- 1 mark: correct definition + matched example for qualitative (words/description; e.g. interview comment).

Q2. Identify · 3 marks

StepUp's developers want to understand why many students stop using the app after two weeks. Identify the most appropriate type of data, name one suitable collection method, and justify your choice.

✗ A weak answer

They should use a survey to get quantitative data so they can make a graph of how many people stopped. Misses the purpose. Counting how many stopped doesn't explain WHY. Wrong type chosen for the stated goal. ~1/3.

✓ A strong answer

They should collect qualitative data, using interviews (or open-ended survey questions) with students who stopped. This is appropriate because the purpose is to understand reasons and motivations — the "why" behind the behaviour — and qualitative data captures rich, detailed explanations that a number could not provide. Type, method and a justification explicitly linked to the purpose ("why"). All three marks.

How the marks are awarded

- 1 mark: identifies qualitative data.
- 1 mark: names a suitable method (interview / open-ended question / focus group).
- 1 mark: justification links the type to the purpose of understanding reasons/the "why".

Q3. Describe · 4 marks

A StepUp survey contains two questions: (i) "On a scale of 1-5, how motivated do you feel to exercise?" (ii) "Describe one thing that stops you exercising." For each question, classify the data collected as qualitative or quantitative and state the purpose it serves.

✗ A weak answer

Both are qualitative because they are about exercise and feelings. The purpose is to help students. Question (i) is wrong — a 1-5 rating is a number, so quantitative. Purposes are vague. ~1/4.

✓ A strong answer

Question (i) collects quantitative data: the response is a number (1-5) that can be averaged and compared across students, so its purpose is to measure and compare motivation levels at scale. Question (ii) collects qualitative data: it is an open response in the student's own words, so its purpose is to understand the specific reasons and barriers behind low exercise. Both classified correctly (note the Likert trap on (i)) with a distinct, sensible purpose for each. Full marks.

How the marks are awarded

- 1 mark: (i) quantitative.
- 1 mark: (i) purpose — measure/compare at scale.
- 1 mark: (ii) qualitative.
- 1 mark: (ii) purpose — understand reasons/barriers.

Q4. Discuss · 6 marks

StepUp collects sensor data (step counts), demographic data (year level) and AI-generated summaries. Discuss how combining quantitative and qualitative data could give the team a more complete understanding of student wellbeing than either type alone. Refer to purpose, and identify one risk the team should manage. ()

✗ A weak answer

Quantitative is good because numbers are facts and qualitative is good because words explain things. Using both is better. AI might be wrong sometimes. Gestures at the right ideas but no development, no worked example, purposes not explained, risk barely addressed. ~2/6.

✓ A strong answer

Quantitative data such as step counts and year-level demographics lets the team measure and compare wellbeing objectively — for example, charting whether activity falls in certain year levels or over time. However, numbers alone can't explain the cause. Qualitative data, such as interview comments or open survey responses, captures the reasons behind the trend — for instance, that a drop in steps was caused by an unsafe winter route. Used together, the quantitative data shows what is happening and how much, while the qualitative data explains why, which supports a better-justified decision about how to improve the app. One risk to manage is the AI-generated summaries: because they are produced by an algorithm rather than observed, they may be biased or inaccurate, so the team should verify any AI insight against the original data before acting on it. Develops both types with worked StepU

How the marks are awarded

- 1-2 marks: explains quantitative purpose with example (measure/compare — what & how much).
- 1-2 marks: explains qualitative purpose with example (explain reasons — why).
- 1 mark: states the benefit of combining (complete picture / better decision).
- 1 mark: identifies and manages a relevant risk (e.g. AI-generated data must be verified for bias/accuracy).